

MD. ASHIQUR RAHMAN HRIDOY

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EDUCATION

Chittagong University of Engineering and Technology (CUET)

BSc in Mechatronics and Industrial Engineering

CGPA: **3.84/4.00**

Merit position: **4th**

Chattogram, Bangladesh

Feb 2020 — Jul 2025

Coursework: Industrial Automation, Robotics, Control System, Sensor and Instrumentation, Computer Integrated Manufacturing (CIM), Machine Learning Algorithm, Operation Research

Achievement:

- Selected for **Dean's list Award** in 2023, 2024, 2025 for achieving CGPA 3.75 or above.
- Technical Scholarship** awarded in all semesters for academic excellence.

RESEARCH INTERSESTS

Industrial Automation | Digital Twin | System Modeling and Simulations | Optimization | Machine Learning | Logistics and Supply Chain Optimization | Sustainable System and Green Manufacturing | Ergonomics and Human Factor Engineering

UNDERGRADUATE THESIS

Optimizing Multi-Depot Vehicle Routing: An ABC-GA Hybrid Algorithm and Comparative Analysis.

- Designed and implemented a hybrid algorithm in Python, focusing on efficient customer allocation and route planning across multiple depots.
- Achieved approximately 2% better performance compared to an existing hybrid algorithm.

PUBLICATION

Hridoy, M. A. R., Lubaba, R. F. K., Roy, S., Dhar, D. **Optimizing Multi-Depot Vehicle Routing: An ABC-GA Hybrid Algorithm** Proceedings of the 8th International Conference on Mechanical Engineering and Renewable Energy (ICMERE 2025), Chattogram, Bangladesh (Under review)

TECHNICAL SKILLS

Programming Language: C programming, Python, MATLAB

Libraries: NumPy, Pandas, Matplotlib, OpenCV, Scikit-Learn, TensorFlow

Design/Simulation: SolidWorks, Proteus, Factory I/O, TIA Portal, FluidSIM

Others: Microsoft Office, Latex

EXPERIENCE

Adjunct Lecturer, Department of Electrical and Electronic Engineering
Uttara University, Dhaka

Jan 2026 — Present

Adjunct Lecturer, Department of Textile Engineering
Uttara University, Dhaka

Sep 2025 — Dec 2025

Responsibilities:

- Preparing and conducting lectures and lab sessions for undergraduate students
- Preparing question papers, evaluating students' performance and assigning grades
- Coordinating with faculty members for course planning and student assessment

Courses Taught:

Theory: Elements of Mechanical Engineering and Machine Design, Mechatronics and Automatic Control Engineering, Signal and System, Industrial and Operations Management

Sessional: Elements of Mechanical Engineering and Machine Design Lab, Workshop Practice

Industrial Training | Walton Hi-Tech Industries PLC

Jan 2025

- Completed a 14-day industrial attachment at Walton Hi-Tech Industries PLC, gaining hands-on exposure to large-scale manufacturing processes including injection molding, thermoforming, and PCB/cable production.
- Observed and analyzed assembly line automation, quality control testing procedures, and industrial workflows for refrigerators and ACs in a high-volume production environment.

PROJECTS

Automatic Coffee Vending Machine using PLC Simulation

Traffic Light Simulation Using PLC S7 1200 in TIA Portal

Line Following Robot

- Developed a car that follows line using IR sensor and Arduino
- Programmed Arduino, integrating real-time sensor feedback with motor control.

Soccer Bot

- Designed and developed a soccer-playing robot controlled via Bluetooth using an Arduino micro-controller.

PROFESSIONAL CERTIFICATION

Certified SolidWorks Associate (CSWA)

Sep 2024

Certified SolidWorks Additive Manufacturing Associate

Oct 2024

Certified SolidWorks Electrical Design Associate

Oct 2024

Road map to ISO 14001:2015 and ISO 45001:2018 Certification

Dec 2023

TRAINING

Industrial Automation | IICT, CUET & BCC, ICT Division

Oct 2024— Jan 2025

- Completed a 60-hour hands-on course in Industrial Automation, covering FluidSIM, TIA Portal, pneumatic and electro pneumatic systems, PLC ladder logic programming, and HMI design

Basic Safety Training | Samsung C&T Corporation, Dhaka

Jul 2023

- A daylong training on First Aid, Fire safety, Personal survival techniques (PST), personal safety and social responsibilities (PSSR), Security Awareness and Emergency procedures.

EXTRACURRICULAR ACTIVITIES

- **Champion**, Techday 2022, An inter university LFR competition organized by RMA
- **Second Runner Up**, Industrial Techathon 2025, A National competition organized by Department of MIE, CUET
- **Qualified**, for pre-final round in International Astronomy and Astrophysics Competition (IAAC) in 2025
- **Participant**: Hult prize at CUET, RMA INSIGHT Technical case solving competition in 2020, RMA Techday 2021 & 2022, Chittagong Science Carnival 2.0, BEC ROBO MANIA 2022 at BUBT
- **Competition Lead**, MIE Robolution 1.0 national event
- **Project Secretary**, Robo-mechatronics Association (RMA)
- **Member**: Industrial Engineering and Operations Management, CUET Chapter

REFERENCES

1. **Homayun Kabir**
Associate Professor, Department of Mechatronics and Industrial Engineering, CUET
homayun@cuet.ac.bd
2. **Sanjeeb Roy**
Assistant Professor, Department of Mechatronics and Industrial Engineering, CUET
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